Problems

1. For the unit place, we need a multiple of 8 whose unit digit is 0. The smallest possiblity is $8 \cdot 5 = 40$. So we must need 5 numbers made of digits 8 to get the total sum as 1000. Similarly for tenth place, we need $4 + 8 \cdot 2 = 20$. For hundredth place, we need $2 + 8 \cdot 1 = 10$. The thousandth place is 1.
   Hence, the final equality is $8 + 8 + 8 + 88 + 888 = 1000$.

2. We know that the sum of the greatest single digit numbers is 18 so the left most digit i.e. A must be 1 so we have 111 + BBB = 111C. So as the rightmost digit is not the same as the other digits in the sum, BBB must be 999 as any other number i.e. 1 to 8 will give us same digits in the sum i.e. DDDD but it is not so.
   Hence, the final equality is 111 + 999 = 1110.

3. $1001 \cdot N = (1000 + 1)N$ where N is any number $(1001 \cdot 715 = 715715)$.

4. $101010101 \cdot 57 = 5757575757$.

5. $10001 \cdot 1020304050 = 10204060804050$.

6. $11111 \cdot 1111 = 12344321$.

7. The equality is given by $1vwxyz \cdot 3 = vwxyz1$. So $3z$ must result in unit digit to be 1 so $z = 7, 14, ...$ but only 7 is appropriate here. Repeating this reveals that our original number is 142857.

8. $123123123 \div 123 = 100100100$.

9. Because our divisor is 7 digits long and our dividend is 100 digits long, the first 98 digits are completely divisible by our divisor and hence 11 is our remainder.

10. $10^{20} \div 7 = 14285714285714285714$ remainder 2.

11. We observe that $1 \div 7 = 0.14285714....$ as 7 cannot perfectly divide 1,2,3,4,5 and 6. This is not a coincidence and arises from the property of the number 7.

12. Following from above we see that

    - $1/7 = 0.142857...$
    - $2/7 = 0.285714...$
    - $3/7 = 0.428571...$
    - $4/7 = 0.571428...$
    - $5/7 = 0.714285...$
    - $6/7 = 0.857142...$

    This happens because 7 is prime and hence it cycles through the remainders when divided by all natural numbers less than 7 i.e. 1,2,3,4,5 and 6.

13. Let 142857 be the first 6 digits of of $\frac{1}{7} \cdot 10^6$. Now multiplying it 1,2,3,4,5,6,7 can be followed from the above. Respectively we get 142857, 285714, 428571, 571428, 714285, 857142 and 1000000.

14. The number 7 is called a full period prime and 142857 is the repetand. Full period primes upto 100 can be represented by the following set (well-defined collection of distinct objects treated as a single entity and do not violate ZFC), $\mathbb{P} = \{7, 17, 19, 23, 29, 47, 59, 61, 97\}$.

15. The generating rule/algorithm:

    - Check the rightmost digit. It may be 0 or 1.
    - If it is 0 then convert it to 1.
    - If it is 1 then convert it to 0 and then turn the left digit 1.
    - If all digits are 1 then convert all digits to 0 and add a 1 to the left of the string of zeroes.

    So the next lines are 1101, 1110, 1111, 10000, 10001, 10010, 10011, 10100, 10101, 10110, ... etc.

16. Yes it is possible to get any number, not just 0 to 31, using the powers of $2 = \{1, 2, 4, 8, 16, 32, ...\}$. You may check any number to verify this claim. This can be proven by the fact that $2^0 = 1$ is valid element in the set of powers of 2 and can represent any natural number on the number line. The other numbers replace 1 when they can be represented by a bigger number.

17. 14 can be written as 1110 because $2^3 + 2^2 + 2^1 = 14$.

18. 45 can be written as 101101.

19. 173 is the number after not bombing basic arithmetic.

20. The following sums can be motivated by thinking of each place from left as a power of 2 and a on for 1 and off for 0. So 0 becomes 1 because you are adding that power and "turning it on". If it was already "on" then you turn it off and turn "on" other appropriate places. This is good exercise.

    - 1010 + 101 = 1111
    - 1111 + 1 = 10000
    - 1011 + 1 = 1100
    - 1111 + 1111 = 11110

21. Same here.

    - 1101 - 101 = 1000
    - 110 - 1 = 101
    - 1000 - 1 = 111

22. This is nice application of the things learnt above. $1101 \cdot 1010 = 1000010$.

23. Lowkey messed up the first time but this is good exercise. $11011 \div 101 = 101$ rem 10.

24. This is twister the first time around. If you know a bit about geometric series then you could find that $\frac{1}{4} + \frac{1}{16} + \frac{1}{64} + ... = \frac{1}{3} \implies \frac{1}{3} = 0.010101....$ If you don't then we do the following.

    - We know that our resulting binary number can be represented as $x = 0.b_1b_2b_3b_4.....$
    - Multiply $\frac{1}{3}$ with 2 and we get $\frac{2}{3} = 0 + \frac{2}{3} = 2x = b_1.b_2b_3b_4...$ so b_1 (the whole part only) must be 0. Hence we have $x = 0.0b_2b_3b_4.....$
    - Multiply $\frac{2}{3}$ with 2 and we get $\frac{4}{3} = 1 + \frac{1}{3} = 4x = b_1b_2.b_3b_4b_5...$ so b_2 (the whole part only) must be 1. Hence we have $x = 0.01b_3b_4b_5....$
    - Repeating the above we get $x = 0.01010101....$

Remember that in base 10 multiplying by 10 moves decimal point to right by one place and in base 2 multiplying by 2 moves decimal point to right by one place.

25. I am craving a coffee so bad but I can't consume caffeine. I guess I'll go watch youtube after this.

26. Calculators never looked better.

27. Calculators: "Nah, I'd win."

28. Calculators: "Ah shit, here we go again."

29. Calculators: "Wait, how can I speak. Stop that. Stop!"

30. Calculators: "A man chooses, a slave obeys I guess."

31. There are 14 ways to do this. You may verify it yourself. The Catalan number corresponding to this setup is 14.

32. 98 pairs of parenthesis is required. Because order of operation can only be defined for more than two distinct numbers and each number after that requires a pair to define order of operation. If N is the total number of distinct numbers then total parenthesis required is $N-2$.

33. $1+2+3+...+50+51+....+98+99+100 = (1+100)+(2+99)+(3+98)+...+(50+51) = 50 \cdot 101 = 5050$.
Let $\mathbb{N}$ be the number of natural numbers, then the sum of natural numbers is given by

$$\frac{\mathbb{N}(\mathbb{N}+1)}{2}$$

.

34. $1001 \cdot 20 = (1000+1) \cdot 20 = 20020$.

35. $1001 \cdot 102 = 102102$.

36. There would be 15 unless you are sleep deprived.

37. The big vessel is 3 liters and the small is 2 liters.

38. Following the order of operations and letting x be any natural number then $(((2(x+3))-x)-4)-x = 2x+6-4-2x = 2$. This shows that the result of the sequence of operations on x is independent of x hence we get 2 for any value of x.

39.